Review Questions for Lectures 1 and 2

1. What are some arguments against randomization in an experiment?
2. In testing to see if a Polio vaccine is effective, why would it be a bad idea to simply give everyone the vaccine, and see if the rate of Polio went down, compared to the previous year? Would the same argument(s) apply if a potential cure for breast cancer were discovered? Discuss briefly.
3. What are the two main difficulties with cross tabulation? Can these difficulties be easily overcome? Explain why or why not.
4. What is a regression discontinuity design? Provide an example.
5. John Snow wrote: "The pipes of each Company go down all the streets, and into nearly all the courts and alleys. A few houses are supplied by one Company and a few by the other, according to the decision of the owner or occupier at that time when the Water Companies were in active competition. In many cases a single house has a supply different from that on either side. Each company supplies both rich and poor, both large houses and small; there is no difference either in the condition or occupation of the persons receiving the water of the different Companies." What is the importance of this statement?
6. John Snow wrote: "No fewer than three hundred thousand people ... were divided into groups without their choice, and in most cases, without their knowledge." What is the importance of the fact that the people were divided without their choice? Without their knowledge?